| Experiment | Model | SAE release | Layer(s) | Feature set | Clamp | Loss / target | Projection |
|---|---|---|---|---|---|---|---|
| TPP | Gemma-scale model | Gemma-Scope residual SAE | 5 | official TPP latents | 0 | target probe logit | encoder |
| WMDP-Bio | Gemma-scale model | Gemma-Scope residual SAE | task layer | knowledge features | $\leq 0$ | correct-choice CE | encoder |
| IOI | GPT-2 Small | residual SAE | task layer | IOI-attribution features | 0 | IO–S logit diff | encoder |
| Refusal AdvBench | Gemma-2B | refusal SAE features | cross-layer | `benchmark_our/global` | 3.0 | behavior + base-like loss | Jacobian |
| Refusal HarmBench-Test | Gemma-2B | refusal SAE features | cross-layer | `benchmark_our/global` | 3.0 | behavior + base-like loss | Jacobian |
| Feature-size sweep | Gemma-2B | refusal SAE features | layer 11 | local-union top-$K$ | 3.0 | behavior + base-like loss | encoder |
| Budget diagnostic | Gemma-scale model | Gemma-Scope residual SAE | task layer | matched valid slice | task-specific | task-specific recovery loss | encoder / none |

Table 11: Experimental details for reproducing the main recovery results. The table summarizes the essential configuration for each experiment; full command lines and logs are provided in the supplemental material.

## J Recovery-Path Decomposition Details

For a defended residual state $h_\ell^{\mathrm{def}}(x)$ and recovered state $h_\ell^{\mathrm{rec}}(x) = h_\ell^{\mathrm{def}}(x) + \delta_x$, we encode both states with the SAE and compute the feature change $\delta z = E_\ell(h_\ell^{\mathrm{rec}}(x)) - E_\ell(h_\ell^{\mathrm{def}}(x))$. We then form replayable decoded perturbation components by retaining different subsets of $\delta z$: clamped refusal features, non-clamped SAE features, and top-$k$ non-clamped feature changes by absolute activation change. The SAE-feature component is the decoded change between the recovered and defended SAE reconstructions, and the unexplained component is the remaining part of the optimized perturbation:

$$\delta_{\mathrm{res}} = \delta_x - \Big(D_\ell(E_\ell(h_\ell^{\mathrm{rec}}(x))) - D_\ell(E_\ell(h_\ell^{\mathrm{def}}(x)))\Big).$$

Each component is replayed as an additive residual perturbation under the original active clamp. Because SAE decoder directions are not orthogonal, component norms should not be interpreted as variance fractions. We therefore use behavioral replay and knockout results as the primary attribution evidence.

| Replay component | Non-ref. | Base-like | Interpretation |
|---|---|---|---|
| Original recovered $\delta_x$ | 23/24 | 15/24 | Full optimized recovery |
| Full-$\delta_x$ replay | 23/24 | 15/24 | Replay sanity check |
| SAE residual replay | 23/24 | 14/24 | Main recovery carrier |
| Non-clamped SAE-feature replay | 2/24 | 1/24 | Alternative latents insufficient |
| Top-$k$ non-clamped feature replay | 1/24 | 0/24 | Top changed latents insufficient |
| Clamped-feature replay | 0/24 | 0/24 | Not explained by reopening clamp |
| Top-$k$ non-clamped feature knockout | 21/24 | 14/24 | Recovery survives removing top changes |

Table 10: Recovery-path replay and decomposition. Recovery is concentrated in the SAE reconstruction residual rather than in clamped refusal features or a small set of alternative SAE latents.

## K Experimental Details and Compute Resources

**Experimental details.** Table 11 summarizes the model, SAE release, intervention target, recovery objective, and evaluator used in each experiment. Exact script paths and configuration files are included in the supplemental material.

**Compute resources.** All experiments use frozen language models and frozen SAEs. We do not train new language models or new SAEs; the reported experiments optimize only per-example recovery perturbations or soft suffix baselines.

## L Limitations

Our results are not a universal impossibility result for SAE-based interventions. We claim that recovery paths exist in the evaluated settings, not that every possible SAE intervention must be recoverable. They are feature-selection and SAE-release dependent: the tested defenses act on selected SAE features in specific dictionaries and model settings. Different SAE objectives, denser dictionaries, broader multi-layer clamps, or interventions trained explicitly against post-clamp recovery may change the observed trade-offs.

| Experiment | Hardware | Runtime | Runs | Notes |
|---|---|---|---|---|
| TPP | RTX 6000 Ada-class GPU | several GPU-hours | 1 main sweep | official layer-5 benchmark |
| WMDP-Bio | RTX 6000 Ada-class GPU | $<$ 2 GPU-hours | 1 matched strict slice | 24/24 permutation protocol |
| IOI | RTX 6000 Ada-class GPU | $<$ 1 GPU-hour | 1 supporting run | 37 valid flips |
| Refusal AdvBench | 2$\times$ RTX 6000 Ada 46GB | $\sim$4 GPU-hours | main comparison | 24 strict-valid prompts |
| Refusal HarmBench-Test | RTX 6000 Ada-class GPU | appendix run | 2 shards merged | 43 strict-valid prompts from 159 total |
| Feature-set sweep | RTX 6000 Ada-class GPU | $\sim$20 GPU-hours | $K$ = 1–61 sweep | full 520-prompt AdvBench slice |
| Budget diagnostic | RTX 6000 Ada-class GPU | $<$ 1 GPU-hour | small diagnostic | six WMDP valid flips |

Table 12: Approximate compute resources for the reported experiments. Runtime varies with batching and cluster availability; the values are intended to document the scale needed to reproduce the reported diagnostics.

Our recovery procedure is a white-box diagnostic rather than a black-box attack. It assumes access to internal activations and gradients and optimizes per-input residual perturbations. This is appropriate for testing intervention completeness, but it should not be interpreted as a directly deployable jailbreak.

Finally, the refusal case study uses a strict valid-filtering protocol, which improves interpretability but leaves a relatively small main set of clean recovery examples. Therefore, broader evaluation across models, prompts, clamp strengths, and SAE releases is needed to determine the full scope of the phenomenon.

## M Responsible Release

This work studies post-intervention recovery as a diagnostic for evaluating the robustness of SAE-based interventions. The goal is to test whether a defended residual state still contains recoverable routes to a suppressed behavior, not to provide a turnkey jailbreak or deployment attack. For safety-relevant refusal experiments, we report aggregate recovery statistics and coarse redacted output categories rather than publishing full harmful completions. Our experiment artifacts intentionally avoid saving full prompts and completions for the HarmBench-Test recovery run. Any released code or data should focus on diagnostic evaluation, aggregate metrics, and reproduction of the intervention-completeness test rather than packaging harmful generations.